

ITSY-1300 Fundamentals of Information Security

TARRANT COUNTY COLLEGE DISTRICT MASTER SYLLABUS

COURSE DESCRIPTION

An introduction to information security including vocabulary and terminology, ethics, the legal environment, and risk management. Identification of exposures and vulnerabilities and appropriate countermeasures are addressed. The importance of appropriate planning, policies and controls is also discussed. Basic information security goals of availability, integrity, accuracy, and confidentiality. Vocabulary and terminology specific to the field of information security are discussed.

COURSE TYPE

Technical

COURSE GOALS AND LEARNING OUTCOMES

End-of-Course Outcomes: Outline best practices for the information security goals of confidentiality, integrity and availability; explain ethical practices; define vocabulary/terminology related to information security; explain the importance of planning and administrative controls; identify security threats, vulnerabilities, and countermeasures; and identify procedures for security risk management.

Learning outcomes are linked to SCANS requirements. Specific SCANS requirements are linked to the class schedule.

Upon successful completion of this course, students will:

1. Demonstrate an understanding of best practices for the information security goals of confidentiality, integrity, and availability by
 - a. contrasting symmetric encryption and asymmetric (public key) encryption (1a, 1b, 2c, 2d, 2e);
 - b. comparing current encryption algorithms (1a, 1b, 2c, 2d, 2e);
 - c. explaining security awareness programs that enable end users to understand their part in information security (1a, 1b, 2b, 2c, 2d, 2e);
 - d. explaining countermeasure best practices (1a, 1b, 2c, 2d, 2e);
 - e. explaining issues related to access control (1a, 1b, 2c, 2d, 2e);
 - f. evaluating software tools to help manage information security (1a, 1b, 2a, 2b, 2c, 2d, 2e).
2. Demonstrate an understanding of ethical practices by
 - a. explaining legal requirements associated with information security, including privacy and computer forensics (1a, 1b, 1c, 2a, 2b, 2c, 2d, 2e);
 - b. demonstrating basic computer forensic processes (1a, 1b, 1c, 2a, 2b, 2c, 2d, 2e).

3. Demonstrate an understanding of vocabulary/terminology related to information security by
 - a. comparing confidentiality, integrity, availability, authentication, and nonrepudiation (1a, 1b, 2c, 2d, 2e);
 - b. explaining the operational model of computer security (1a, 1b, 2c, 2d, 2e);
 - c. explaining how various security technologies and techniques fit within the operational model (1a, 1b, 2c, 2d, 2e);
 - d. explaining vocabulary related to encryption and public key infrastructure (1a, 1b, 2c, 2d, 2e).
4. Demonstrate an understanding of the importance of planning and administrative controls by
 - a. explaining the impact of IT and software engineering project management and change management on information security (1a, 1b, 2a, 2b, 2c, 2d, 2e);
 - b. demonstrating the ability to explain the importance of and to apply the concepts of least privilege and need to know in information security (1a, 1b, 1c, 2a, 2b, 2c, 2d, 2e);
 - c. explaining the importance of disaster recovery and business continuity plans (1a, 1b, 2a, 2b, 2c, 2d, 2e);
 - d. identifying best practices for disaster recovery and business continuity plans (1a, 1b, 2a, 2b, 2c, 2d, 2e).
5. Demonstrate an understanding of security threats, vulnerabilities, and countermeasures by
 - a. comparing vulnerabilities and threats (1a, 1b, 2c, 2d, 2e);
 - b. explaining common networking vulnerabilities, including wired and wireless networks (1a, 1b, 2c, 2d, 2e);
 - c. identifying common malicious, environmental, and physical threats and the countermeasures used to combat them (1a, 1b, 2a, 2b, 2c, 2d, 2e);
 - d. identifying unintentional threats and the countermeasures used to combat them (1a, 1b, 2a, 2b, 2c, 2d, 2e).
6. Demonstrate an understanding of procedures for security risk management by
 - a. explaining the importance of risk management to information security (1a, 1b, 2a, 2b, 2c, 2d, 2e);
 - b. comparing qualitative and quantitative risk assessment (1a, 1b, 2a, 2b, 2c, 2d, 2e);
 - c. explaining tools used in risk management (1a, 1b, 2a, 2b, 2c, 2d, 2e).

COURSE ASSESSMENT

Student success is measured by a variety of assessment techniques aligned with course goals and learning outcomes. Individual faculty members are responsible for designing evaluation instruments to measure student mastery of course goals and learning outcomes and indicating the nature of such evaluation instruments in the instructor's class requirements.

STUDENT ACCESSIBILITY RESOURCES

Any student with a documented disability needing academic accommodations is required to contact the Student Accessibility Resources (SAR) Office located on each campus to schedule an appointment with the Coordinator of SAR. All discussions are confidential. Because SAR accommodations may require early planning and are not provided retroactively, students are encouraged to contact SAR as early in the semester as possible. SAR is responsible for approving and coordinating all disability-related services. TCC professors will honor requests for accommodation when they are issued by SAR.

SCHOLASTIC DISHONESTY

Students are responsible for adhering to the TCCD policy on scholastic dishonesty as stated in the online student handbook at www.tccd.edu.

RESOURCES

For additional information regarding the student handbook, academic calendar, course evaluations, attendance policy, SCANS skills, Core Competencies, etc., see www.tccd.edu.